Residue stabilizers and distinguished generators

Human Compression project

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This manuscript supplies the general arguments for the intake of Fernandes–Leemans–Weiss, *Highly symmetric hypertopes* (2016). Its central observation is that a stabilizer of a flag can be read from paths of chambers in its residue. Ordinary paths give the distinguished involutions of a regular hypertope; even paths give the distinguished generators of a chiral hypertope. Intersections then follow from intersections of pointwise stabilizers. We use the already acquired propagation, extension-fiber and reflection theorems in the chamber manuscript. Their proofs are dependencies, not new assertions of the present application. Finite examples, numerical tables and the toroidal families are separate components.

1 Conventions and acquired geometric arguments

Let I be a finite type set. An incidence geometry has one object of each type in every chamber, and every flag extends to a chamber. Distinct objects of the same type are not incident. The residue of a flag F consists of the objects of the remaining types incident with every object of F. Residual connectedness means incidence connectedness of every residue of rank at least two, including the empty-flag residue. Thinness means that every panel is contained in exactly two chambers.

For a thin geometry write C^i for the unique chamber different from C on the panel obtained by deleting its type-i object. Superscripts are applied in their written order: $C^{ij} = (C^i)^j$. All group actions in this manuscript are *left* actions; thus $(gh)C = g(hC)$. An automorphism preserves types unless stated otherwise. It commutes with every operation $C \mapsto C^i$.

The acquired propagation and reconstruction lemma has three conclusions:

1. Chambers containing a fixed flag are connected by adjacency steps which retain that flag.
2. In a thin residually connected geometry, an automorphism fixing a chamber is the identity.
3. A permutation of chambers preserving every adjacency color comes from a unique automorphism of the incidence geometry.

The proof of the first conclusion is an induction on the rank of the residue. It is important to apply the induction to residues known to be connected. Mere incidence connectedness of the whole geometry does not supply that hypothesis. The converse needed for Proposition 2.1 of the source is elementary: a chamber path in a residue of rank at least two gives an incidence path between any two of its objects. Consecutive chambers share at least one object, and each object lies in a chamber. Thus residual connectedness and strong chamber connectedness are equivalent under the source's hypotheses. This corrects an intermediate step in its proof without changing the stated proposition.

Chamber transitivity implies transitivity on flags of every fixed type: extend both flags to chambers and carry one chamber to the other. Conversely chamber transitivity is a special case of flag transitivity. This is Proposition 2.2 of the source.

2 The residue stabilizer identity

Fix a chamber $C = \{c_i : i \in I\}$ and put

$$F_{I\setminus S} = \{c_i : i \notin S\} \qquad (S \subseteq I).$$

Regularity means transitivity of the full automorphism group on chambers; by propagation its action is free as well. The same conclusions below hold for any chamber-transitive group of actual automorphisms, which then equals the full group by freeness.

Theorem 2.1. *Let G act regularly on the chambers of a hypertope. There is a unique $\rho_i \in G$ with $\rho_i C = C^i$ for each $i \in I$. These elements are distinct nonidentity involutions. For every $S \subseteq I$,*

$$\langle \rho_i : i \in S\rangle = \mathrm{Stab}_G(F_{I\setminus S}). \tag{1}$$

In particular they generate G and satisfy the full intersection property

$$\langle \rho_i : i \in S\rangle \cap \langle \rho_i : i \in T\rangle = \langle \rho_i : i \in S \cap T\rangle.$$

Proof. Existence and uniqueness of ρ_i follow from regularity. Since $\rho_i(C^i) = (\rho_i C)^i = C$, its square fixes C, so $\rho_i^2 = 1$. It is nonidentity because $C^i \neq C$. The chambers C^i, C^j are distinct when $i \neq j$, so the generators are distinct. Each ρ_i fixes every c_j with $j \neq i$. This proves inclusion from left to right in (1).

Let g fix $F_{I\setminus S}$. The chambers C and gC contain this flag. Propagation inside its residue gives a path from C to gC whose colors all belong to S. If its colors are $i_1, \ldots, i_m$, its successive chambers are

$$C, \quad \rho_{i_1}C, \quad \rho_{i_1}\rho_{i_2}C, \quad \ldots, \quad \rho_{i_1}\cdots\rho_{i_m}C.$$

Indeed adjacency commutes with every group element. The last product and g have the same image of C, and freeness makes them equal. This also works for $S = \varnothing$, when the stabilizer is trivial. It proves (1). Taking $S = I$ proves generation. Finally, to fix both complementary flags is exactly to fix their union, which is the complementary flag of $S \cap T$. Applying the identity three times gives the claimed intersection. □

This proves source Theorem 4.1 without an induction that presumes the identification of the residue group. Its rank-two Coxeter quotient interpretation follows as well: put $m_{ij} = o(\rho_i\rho_j)$, allowing ∞. The Coxeter presentation with these labels maps onto G. The rank-two subgroups are dihedral of order $2m_{ij}$ (infinite when the label is infinite), so the map is injective on each such subgroup.

For the chiral case, $G = \mathrm{Aut}_I(\Gamma)$ has exactly two chamber orbits, and adjacent chambers lie in different orbits. Call the orbit of C positive. Every two-step chamber C^{ij} is positive. The action is free by propagation, so there is a unique element taking C to C^{ij}. Fix $0 \in I$, put $\alpha_0 = 1$, and for $i \neq 0$ define $\alpha_i C = C^{0i}$.

Theorem 2.2. *For a chiral hypertope of rank at least three set*

$$\alpha_{ij} = \alpha_i^{-1}\alpha_j, \qquad H_S = \langle \alpha_{ij} : i, j \in S\rangle \quad (S \subseteq I).$$

Then $\alpha_{ij}C = C^{ij}$ and

$$H_S = \mathrm{Stab}_G(F_{I\setminus S}). \tag{2}$$

Consequently $G = \langle \alpha_i : i \neq 0\rangle$ and $H_S \cap H_T = H_{S\cap T}$ for all subsets S, T. The set of nonidentity distinguished generators $\{\alpha_i : i \neq 0\}$ is independent.

Proof. First $\alpha_i^{-1}C = C^{i0}$, since applying α_i to the latter chamber gives $C^{0ii0} = C$. Therefore

$$\alpha_i^{-1}\alpha_j C = \alpha_i^{-1}(C^{0j}) = (C^{i0})^{0j} = C^{ij}.$$

The same identity includes $i = 0$, $j = 0$ and $i = j$. In particular α_{ij} fixes every object of C outside types i, j, so H_S is contained in the stabilizer in (2).

For the reverse inclusion let g fix the complementary flag. Propagation gives a path from C to gC using colors in S. Every adjacency changes the chamber orbit. Since both endpoints are positive, the path has even length, say with colors $i_1, j_1, \ldots, i_m, j_m$. After its first pair the chamber is $\alpha_{i_1 j_1} C$. Equivariance then shows successively that its last chamber is $\alpha_{i_1 j_1} \cdots \alpha_{i_m j_m} C$. Freeness makes this product equal to g.

This argument includes singleton and empty S. Their group H_S is trivial; equivalently a group element stabilizing a panel cannot exchange its two chambers because those chambers have different colors. We have proved the stabilizer identity for every S, rather than assumed that a residue has at most two orbits. Generation follows on taking $S = I$; intersections follow by taking the union of the two fixed flags.

The chamber C^{0i} differs from C for $i \neq 0$: equality would imply $C^0 = C^i$, which is impossible for different types. Hence $\alpha_i \neq 1$. The subgroup generated by the other α_j is $H_{I\setminus\{i\}}$, whereas $\alpha_i \in H_{\{0,i\}}$. Their intersection is $H_{\{0\}} = 1$. Thus no α_i lies in the subgroup generated by the others. □

The intersection conclusion contains the source's condition IC^+, which is stated only for $|S|, |T| \geq 2$. This proves Theorem 7.1 and Corollary 7.2. Ranks at most two have no chiral hypertopes: a connected thin rank-two incidence graph is a cycle or a doubly infinite path, whose type-preserving automorphisms are transitive on its edges; the rank-one case is the two-object geometry.

Corollary 2.3. *The chiral group is transitive on flags of every proper type set. In particular every proper truncation is flag-transitive.*

Proof. Let F be a flag of proper type J and extend it to a chamber D. If D is negative, choose $i \notin J$ and replace it by D^i; this retains F and makes the chamber positive. Some $g \in G$ takes C to the resulting positive chamber and therefore takes its J-subflag to F. A flag in a truncation extends to a chamber of that truncation by extending it in the whole geometry. The same transitivity argument applies there. □

This includes Proposition 8.1 of the source. It asserts transitivity, not thinness or regularity of those truncations.

3 Cosets and the converse construction

For subgroups $P_i \leq G$, the objects of type i are left cosets gP_i, with incidence given by nonempty intersection. This always defines an incidence system and has the base chamber $\{P_i : i \in I\}$. It need not have the flag-extension property in rank at least four. Left multiplication is transitive on objects of each type and on incident pairs: take a common representative of the two cosets. These are the precise assertions used from source Proposition 3.1.

The kernel of the action is

$$K = \bigcap_{i \in I} \bigcap_{g \in G} g P_i g^{-1}.$$

It is contained in every P_i. Sending gP_i to $(gK)(P_i/K)$ is a bijection on each type. It preserves incidence in both directions, because a representative of an intersecting quotient coset can be lifted and the discrepancy is in $K \leq P_i$. Thus it gives the isomorphism of source Theorem 3.3. A claim that G itself acts faithfully before this quotient would be false.

If a group is transitive on objects and incident pairs of each type, its geometry is reconstructed by these cosets of the stabilizers of a base chamber. Indeed $gP_i \mapsto gc_i$ is a well-defined bijection; two image objects are incident precisely when the pair is a translate of the base pair, which is equivalent to the two cosets intersecting. This applies to regular hypertopes and, by Corollary 2.3, to chiral hypertopes of rank at least three. The residue stabilizer identities identify their P_i with the prescribed marked subgroups.

For a regular faithful action the Borel subgroup $B = \bigcap_i P_i$ is the chamber stabilizer and hence is trivial. The stabilizer of a panel has exactly two elements, since it acts transitively and freely on the two containing chambers. These observations prove Lemma 3.4 and its stated consequence. For a nonfaithful chamber-transitive coset action on a hypertope, the corresponding conclusion is $B = K$ and a panel stabilizer has quotient of order two by K; passage to the faithful quotient is essential.

Theorem 3.1. *Let $(G, \{\rho_i : i \in I\})$ be a C-group and put $G_S = \langle \rho_i : i \in S \rangle$ and $P_i = G_{I\setminus\{i\}}$. If its coset incidence system is flag-transitive under G, it is a regular hypertope. Its full automorphism group is G.*

Proof. Every flag is a translate of the corresponding base flag, so extends to a chamber. Iterated use of the C-group intersection property gives

$$\bigcap_{j \notin S} P_j = G_S \qquad (S \subseteq I),$$

including the conventions for empty intersections. The base chamber stabilizer is $G_\varnothing = 1$, so the action is faithful and free on chambers. The stabilizer of the complementary base flag acts transitively on the chambers containing that flag: a group element taking the base chamber to any such chamber must fix every object of the flag. For a panel its stabilizer is $G_{\{i\}} = \{1, \rho_i\}$; hence the geometry is thin.

More generally its residue chambers are the chambers gC with $g \in G_S$. A word in the ρ_i, $i \in S$, gives an adjacency path between C and gC wholly inside that residue. Thus all residues are chamber-connected and therefore incidence-connected in rank at least two. The geometry is a hypertope. Its full automorphism group acts freely on chambers, whereas G already acts transitively, so the two groups coincide. □

This proves source Theorem 4.6. It does not assert flag transitivity from the intersection property alone.

4 The classical coset criteria

The following elementary arguments record exactly which group conditions are used. Their origins remain Tits, Dehon and Buekenhout–Hermand as attributed in the source; they are not new claims of priority.

Proposition 4.1. *Suppose the coset incidence system is flag-transitive. Put $P_J = \bigcap_{j\in J} P_j$, with $P_\varnothing = G$. It is residually connected if and only if*

$$P_J = \langle P_{J\cup\{k\}} : k \in I \setminus J \rangle \qquad (|I \setminus J| \geq 2).$$

Proof. Fix the base flag of type J and write $H = P_J$. Every incident object of type $k \notin J$ has the form hP_k with $h \in H$, by flag transitivity on $J \cup \{k\}$. Two such representatives give the same object precisely when they give the same left coset of $H \cap P_k$ in H. If two objects of the residue are incident, flag transitivity on their union with the base flag supplies a common representative in H. Consequently the residue is exactly the coset incidence system of H with subgroups $H \cap P_k$.

In a coset incidence graph of rank at least two, the component containing the base flag is invariant under each of its vertex stabilizers: an automorphism fixing a vertex preserves that vertex's component. It therefore contains every vertex represented by an element of the group L generated by those stabilizers. Conversely, since each stabilizer is contained in L, assigning the label hL to a vertex represented by h is well-defined and constant along edges. The graph is connected if and only if $L = H$. Apply this to every J. □

Without flag transitivity, a residue can have additional incidence edges; the equality with a coset graph must not be assumed.

Proposition 4.2. *For each nonempty $J \subseteq I$ choose $a(J) \in J$. The coset system is flag-transitive if and only if, for every $|J| \geq 3$,*

$$\bigcap_{j \in J \setminus \{a(J)\}} P_j P_{a(J)} = \left(\bigcap_{j \in J \setminus \{a(J)\}} P_j \right) P_{a(J)}.$$

Proof. The acquired extension-fiber theorem identifies orbits on flags of type $T \cup \{k\}$, assuming transitivity on type T, with

$$P_T \backslash \left(\bigcap_{j \in T} P_j P_k \right) / P_k.$$

There is one orbit precisely when the middle set is $P_T P_k$. Transitivity on flags of size at most two always holds. Inducting on $|J|$, with $T = J \setminus \{a(J)\}$, proves sufficiency. Necessity is the same orbit calculation at every step. Every flag then translates to a base flag and extends, so no missing geometry hypothesis is hidden in this test. □

Lemma 4.3. *For subgroups A, B, D the following are equivalent:*

$$AB \cap AD = A(B \cap D), \tag{i}$$

$$(A \cap B)(A \cap D) = A \cap BD, \tag{ii}$$

and (iii): three right cosets of A, B, D that meet pairwise have a common point.

Proof. Suppose (i), and let $x = bd \in A \cap BD$. Then $b = xd^{-1}$ belongs to $AB \cap AD$, so $b = ae$ with $a \in A$ and $e \in B \cap D$. Since $b, e \in B$, we have $a \in A \cap B$; and $ed = a^{-1}x$ belongs to $A \cap D$. This proves the nontrivial inclusion of (ii).

Conversely let $x = ab = cd \in AB \cap AD$, with $a, c \in A$, $b \in B$, $d \in D$. Inverting equality (ii) also gives $A \cap DB = (A \cap D)(A \cap B)$. Thus $c^{-1}a = db^{-1} = uv$, where $u \in A \cap D$ and $v \in A \cap B$. It follows that $x = cu(vb)$, with $vb = u^{-1}d \in B \cap D$. This gives (i); the opposite inclusions in both equalities are automatic.

For the coset assertion, translate a common point of the B- and D-cosets to the identity by right multiplication. The three cosets then have the form Ak, B, D. They meet pairwise precisely when $k \in AB \cap AD$. They have a common point precisely when $k \in A(B \cap D)$. This is exactly (i). □

This is source Lemma 4.2. Inverting cosets gives the equivalent statement for the left-coset convention used elsewhere here.

Corollary 4.4. *For a rank-three C-group the standard coset geometry is thin if and only if the designated group acts regularly. In either case it is a regular hypertope.*

Proof. Any rank-three coset system is a geometry: an incident pair has a common representative, which supplies an object of the third type. If the group is chamber-transitive, Theorem 3.1 applies.

Now suppose it is thin. Normalize any incident pair to P_i, P_j, and let k be the remaining type. There are two distinct completions P_k and $\rho_k P_k$: the element ρ_k belongs to $P_i \cap P_j$ and not to P_k, by the intersection property. Both completed chambers are translates of the base chamber. Thinness allows no other completion. Hence every chamber is a translate of the base chamber, and the theorem applies. No inference of freeness on all chambers from the base stabilizer alone is used. □

5 Rotation subgroups and strings

Proposition 5.1. *Let $(G, \{\rho_i : i \in I\})$ be a C-group with $0 \in I$. Set $\alpha_0 = 1$, $\alpha_i = \rho_0\rho_i$ for $i \neq 0$ and $H = \langle \alpha_i : i \neq 0 \rangle$. These generators are independent. The index $[G : H]$ is one or two. If it is two, then*

$$\langle \alpha_i^{-1}\alpha_j : i, j \in S \rangle = G_S \cap H \qquad (S \subseteq I),$$

so the rotation generators satisfy IC^+.

Proof. Conjugation by ρ_0 inverts every α_i, and $\rho_i = \rho_0\alpha_i$. Thus $H \lhd G$ and G/H is generated by the image of ρ_0, whose order is at most two. If α_i lay in the subgroup generated by the other α_j, it would belong to $G_{I\setminus\{i\}}$. Multiplication by ρ_0 would put ρ_i in that subgroup, contradicting the intersection property. This proves independence regardless of the index.

If the index is two, all ρ_i have the same nontrivial image in G/H. Consequently a word in the ρ_i, $i \in S$, represents an element of H exactly when its length is even. Every such word is a product of pairs $\rho_i\rho_j = \alpha_i^{-1}\alpha_j$ with $i, j \in S$. This proves the displayed identity. Intersecting two such identities and using $G_S \cap G_T = G_{S\cap T}$ proves IC^+. □

These are Propositions 6.1 and 6.2. The index-two hypothesis in the last assertion cannot be discarded; the source's Example 6.3 is checked in the finite component.

For source Theorem 5.1 we retain the standard string C-group correspondence with abstract regular polytopes, in its exact marked form: the faces of type i are the cosets of $G_{I\setminus\{i\}}$, its flag action is regular, and its rank-two sections have polygon sizes $o(\rho_i\rho_j)$. This is the classical Aschbacher–Schulte input cited in that theorem and the McMullen–Schulte correspondence already retained by the programme. It yields flag transitivity, after which Theorem 3.1 gives the hypertope statement. This input is not counted as a new proof here.

Conversely, for a regular hypertope the rank-two residue on types i, j has group $G_{\{i,j\}}$ by Theorem 2.1. Its alternating chamber cycle has length $2o(\rho_i\rho_j)$, or is infinite if that order is infinite. A missing edge in its diagram means this residue is a digon, so the product order is two. If the diagram is a string, every pair of nonconsecutive types is missing, and the distinguished C-group is therefore a string C-group. This proves source Theorem 5.2.

6 The two chamber orbits in Construction 8.1

Let $I = \{0, \ldots, r-1\}$ with $r \geq 3$. Suppose $G = \langle \alpha_1, \ldots, \alpha_{r-1} \rangle$ is a C^+-group with its independent nonidentity generators. Set $\alpha_0 = 1$ and $H_S = \langle \alpha_i^{-1}\alpha_j : i, j \in S \rangle$. The parabolics of Construction 8.1 are

$$P_i = H_{I\setminus\{i\}} \quad (i \in I).$$

For $i > 0$ this is $\langle \alpha_j : j \neq 0, i \rangle$; for $i = 0$ it is $\langle \alpha_1^{-1}\alpha_j : 2 \leq j < r \rangle$, by telescoping the pairwise differences. We continue to assume, as the source theorem does, that the associated coset system is a hypertope and that G has *exactly two* orbits on its chambers.

Iterating IC^+ gives trivial intersections of all parabolics but one: the common support is a singleton and its H-group is trivial. The iteration uses IC^+ while both supports have size at least two; as soon as a singleton is reached the intersection is already trivial. The base chamber stabilizer is trivial as well. Thus this is a faithful action, and propagation makes it free on every chamber.

Write $C = \{P_i : i \in I\}$. The coset $\alpha_1 P_0$ contains every α_j, $j > 0$. For each $i > 0$ choose $j > 0$ different from i; then $\alpha_j \in P_i$, so replacing P_0 by $\alpha_1 P_0$ gives a chamber. It differs from C: otherwise $\alpha_1 \in H_{I\setminus\{0\}} \cap H_{\{0,1\}} = H_{\{1\}} = 1$. By thinness it is C^0; denote it by C_0.

For $i > 0$, the coset $\alpha_i^{-1} P_i$ is different from P_i by independence. It meets P_k for $k > 0$, $k \neq i$, at α_i^{-1}. It meets P_0 at $\alpha_i^{-1}\alpha_j$, where $j > 0$, $j \neq i$. Replacing P_i by this coset therefore gives the chamber C^i. On multiplying it by α_i, every positive-type object becomes the corresponding object of C_0, while $\alpha_i P_0 = \alpha_1 P_0$. Hence

$$C^i = \alpha_i^{-1} C_0 \qquad (i \in I).$$

Here $i = 0$ follows from $\alpha_0 = 1$.

No element of G takes C to any C^i, since it would fix the base panel and its stabilizer is trivial. The same holds at any translate of C. Also every neighbor of a translate of C_0 is a translate of C: this follows from $C^i = \alpha_i^{-1} C_0$ and involutivity of adjacency. With the exactly-two-orbits hypothesis, all chambers are therefore uniquely

$$(g, +) = gC, \qquad (g, -) = gC_0, \qquad (g, +) \sim_i (g\alpha_i^{-1}, -).$$

These are precisely the hypotheses and coordinates of the acquired two-regular-orbits theorem. It follows that the full group is G, and the hypertope is chiral, unless some $\theta \in \mathrm{Aut}(G)$ inverts every α_i. If such a θ exists, the explicit color-preserving map

$$(g, +) \mapsto (\theta(g), -), \qquad (g, -) \mapsto (\theta(g), +)$$

extends to an incidence involution F, and the full group is $G \rtimes \langle F \rangle$, with $FgF^{-1} = \theta(g)$. This proves source Theorem 8.2 by the already acquired theorem. In this notation the adjoined chamber involution can be nontrivial even when the abstract automorphism θ is the identity.

The proof uses the two-orbits hypothesis to identify the entire chamber set. It does not establish that hypothesis for an arbitrary C^+ presentation. Nor does an abstract semidirect product alone establish that its extra involution acts on the proposed geometry.

7 Scope, attribution and source questions

Theorems 2.1 and 2.2 replace the two separate intersection arguments by the same residue-stabilizer identity. The converse construction and the rotation argument are then short deductions. The geometric propagation lemma, extension-fiber theorem and explicit reflection theorem are acquired project proofs, with their earlier attributions unchanged. The subgroup criteria remain the classical results attributed by the source to Tits, Dehon and Buekenhout–Hermand. The string correspondence remains an explicit retained theorem, not an inferred consequence of an order calculation. No claim of historical priority for the stabilizer observation is made.

The programme already has reviewed negative answers to source Problems 9.3 and 9.4 under the standard parabolic construction. Their precise statements are retained here. Thinness and chamber connectedness force the standard C-group orbit to contain all chambers, excluding the example requested in 9.3. Every rank-three standard coset system on three generators is residually connected, excluding the example requested in 9.4. The latter argument only needs a connected *spanning* coset graph in each residue and does not assert an equality of residue incidence relations without flag transitivity. These are acquisitions of earlier answers, not two new solutions generated by this intake.

Problems 9.1 and 9.2 and the conjectured completeness of the known chiral list remain separately registered questions. This component supplies no classification of all locally toroidal cases. It grants no whole-original credit before the finite and family components and the final source map are reviewed. Existing Lean declarations retain exactly their previous algebraic and action scope; no new formalization or compression measurement is asserted here.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Mathematicae 90 (2016), 1045–1067. Fixed accepted author manuscript, 22 pages.

[2] Human Compression project, *Chambers and marked covers*, topic-chamber/v2, sections on normalization, propagation and reconstruction, and two regular orbits. The accepted version and review are acquired by their exact artifact bindings.

[3] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, Section 2E. The marked regular-polytope correspondence is retained, as in the existing programme.